

The Automotive Manufacturing Industry as a Potential Twin Transformer: A Research-In-Progress Perspective

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Abstract: Twin transformation offers an orchestrated solution to the challenges encountered in digitalization and sustainability transformation endeavours. Despite current research efforts of this concept, there is a scarcity of empirical evidence related to its foreseen benefits, its underlying twin transformation maturity, and its implication among the manufacturing sector. We propose the automotive manufacturing industry as a promising case due to its global economic, societal, sustainability and business model relevance. Therefore, we provide theoretical context of twin transformation, its role in the manufacturing sector as well as in the automotive manufacturing industry. Additionally, we pair it with our ongoing research based on an echeloned design science research, including a preliminary problem analysis, and highlight potential future research directions on twin transformation.

Keywords: Twin Transformation, Automotive Manufacturing Industry, Maturity Model, Case Study

1 Introduction

Due to the constant need to stay competitive while, simultaneously, complying with climate-neutral policies, organizations keep their digital and sustainability transformations as a top strategic priorities [Br24]. While on one side, the digital transformation aims to improve organizations and their strategies through the use of digital technologies with a focus on economic value creation [Vi19], on the other side, the sustainability transformation aims to improve organizations through a positive impact on the society and the planet [DM16]. Aiming to avoid a collision of both digital and sustainable interests, the twin transformation idea arose [Gr23].

The twin transformation offers a value creation harmonization of both undertakings where, on one end, the digital transformation entitles the sustainability transformation and, on the other end, the sustainability transformation reshapes the digital transformation [Gr23]. Despite the acceptance of its novelty, including, IT service providers experts [Br24], within the research of the twin transformation, as in its related term “twin transition”, there are still various open questions that scholars ask for attention [TPC25]. In particular, there is

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a need for empirical evidence to support the twin transformation [Ca24], further development of its maturity model [Br24], as well as its further study in the manufacturing sector [Br24; Ca24; TPC25].

In order to expand the body of knowledge related to twin transformation, we propose to take this important step by investigating the automotive industry due to its economic, political, and environment relevance for society [Pi21] as well to its ongoing digital and sustainability transformations [Ce23; LRV21]. Its significance is demonstrated by the fact that alone the automotive industry is responsible for the total of 27 % of the CO2 emissions in the EU [Pi21]. Since the manufacturing activities are the core of its business model [MF10], we advocate to delve into this sector as a research starting point having as a primary goal to expand the twin transformation capability maturity model (TTCMM) by [Br24] within its operations dimension. Thus, we ask the research question: *What is the twin transformation maturity of the automotive manufacturing industry?*

In this paper, we proceed as follows: First, we present an overview of the twin transformation state of the art and deep dive into the TTCMM aiming to bring light to potential research avenues. Second, we present the current status of related research areas to the twin transformation in the manufacturing sector and advocate for an approach to expand the TTCMM in the operations dimension. And third and finally, we present the automotive manufacturing industry as a future use case, highlighting the planned methodology for our ongoing study based on echeloned design science research (eDSR) by [TWV24] including an introduction to the first echelon, problem analysis. This first echelon includes the analysis of one automotive company which provides a preliminary answer to our research question. Overall, within this paper, chapter two and three offer a theoretical background while the third chapter presents our chose methodology for the next steps of our research with an initial conceptual analysis. Although the present research idea is not fully executed or yet implemented, we aim to contribute to the current body of knowledge of twin transformation, that has become increasingly relevant for both practice and academia.

Based on the nature of the topic and related fields, our present work can be initially, but not limited, be referred to the following sustainability development goals of the UN: (9) industry, innovation and infrastructure, (12) responsible consumption and production & (13) climate action [Un25].

2 Twin Transformation

Since the beginning of the decade, the call for an interplay of digital and sustainability transformation has been made searching for an answer to the need of organizations and government to become more competitive, sustainable, and ethical [TPC25; Vi19]. The study of the interaction of these two endeavors has been seen through multiple lenses, including but not limited to twin transition, digitally-enabled sustainable transition, green IT and twin transformation [Gr23; STT24; TPC25]. Twin transformation has emerged interest by practitioners [Br24; Ch24; Gr23] since it presented a promising approach to the

digital-innovation compromise due to its harmonization nature and value proposal. Twin transformation is defined as “value-adding interplay between digital and sustainability transformation efforts that improve an organization by leveraging digital technologies for enabling sustainability and leveraging sustainability for guiding digital progress” [Ch24].

Within the core investigation of the twin transformation, various approaches have been presented with the idea to land the concept, create the basis for theory development and therefore turn on the flame of this promising field [Br24; Ch24; Gr23]. To begin with, a twin transformation model was developed aiming to illustrate the interrelation of digitalization and sustainability inside organizations [Gr23]. Next, a twin transformation capability framework presented dynamic capabilities for organizations interested in twin transform [Ch24]. Furthermore, a TTCMM was crafted to examine maturity of the twin transformation of organizations based according to various maturity stages [Br24]. And finally, a twin transformation resource orchestration pyramid was recently tailored to describe the resource management processes throughout twin transformation [Bu25].

The TTCMM stands out due to the tangible implications to evaluate the twin transformation maturity of organizations, as in digital transformations [Be20], with the help of capabilities and it establishes an initial map to the summit called “true twin transformer” [Br24]. The idea is straightforward, represented with a pyramid whose levels are the maturity stages. The maturity stages are, with a bottom-up perspective: (1) awareness of combining digitalization and sustainability, (2) twin transformation development, (3) twin transformation implementation and (4) true twin transformer [Br24]. The maturity stages are divided into capability dimensions that help to identify in which state the organization is and what actions they should take to move to the next maturity stage, such as for example the operations dimension focused on the value creation processes. Each organization can start and follow its own pathway based on its own maturity of digitalization and sustainability [Br24]. The pathways are: Digital transformation (DT) expert, sustainability transformation (ST) expert, or twin transformation newcomer [Br24].

Despite the novelty presented in the TTCMM, further investigation and implementation is essential such as the further development of the capability dimensions and the application of the maturity model in various sectors [Br24]. The manufacturing sector promises a valuable use case that could fulfill the need for further research and evidence for the operations dimension [ABA25; Br24] due to its focus on value creation, along with providing the required solid empirical evidence to support the twin transformation construct [Br24; Ca24].

In the next chapter we present the state of the art of the twin transformation in the manufacturing sector, a potential improvement to the TTCMM through the operations dimension and an overview of manufacturing sectors.

3 Twin Transformation in the Manufacturing Sector

The manufacturing sector is facing various challenges regarding the implementation and acceleration of its sustainability transformations derived by continuous pressure to reduce their environmental impact, align with the market requests, follow stronger policies and increase its value creation [STT24]. As a solution to this need, the digital transformation has been studied and implemented in the form of industry 4.0 [Re23], circular economy strategies [Uh22], digitally-enabled sustainable transition [Uh22], among others [ABA25]. In the previously mentioned attempts, the sustainability and digitalization interactions are focused on specific topics and not as an orchestrated transformation.

Since there is a lack of literature regarding twin transformation in the manufacturing sector, we expanded our initial literature scope to the term twin transition. With the introduction of twin transition in a first stage, a new perspective was presented in the manufacturing sector. In this regard, research has focused on the identification of skills like resilience and digital, technical expertise [Ho24], capabilities identification in the value creation processes [Kü24], and the development of frameworks for manufacturing systems [VR23] to mention a few.

The design and operations (DesOps) framework presented by [VR23] aims to improve the efficiency of manufacturing systems based on the idea of twin transition. It represents major progress in this area since, currently, manufacturers do not accelerate their digital transformation due to a more conservative mindset [Ca18]. At this point, we would like to focus on the maturity models considered within the DesOps framework since they contribute to our research question.

The maturity models are divided into three domains, digitalization, automation, and circularity, and include the latest maturity models for manufacturing systems [VR23]. First, the digitalization domain focuses on the digitalization selection for manufacturing companies [Sc20]. Second, the automation domain specializes on the autonomy level with an artificial intelligence perspective [Ah19]. And third, the circularity domain takes care of the measurement of the circular economy [Ac21]. Based on its impact and relevance for the automotive industry [Ce23], the maturity models for manufacturing systems presented in the DesOps framework propose a practical solution to the need for further research in the operations dimension of the TTCMM [Br24].

In various sectors within the manufacturing sector, the twin transition has been studied mostly based on qualitative research [ABA25], covering from small and medium enterprises [ABA25], regions and products [TPC25] and specialized industries such as: farming [Da24], cosmetics [Pe23]. Based on an initial review, empirical research has not been made in relation to the twin transformation in the automotive industry. This sector could provide promising data due to its economic impact, size and regionality [Ca24; LRV21; Pi21].

In the next chapter, we present the status of the automotive manufacturing industry in

relation to twin transformation, its maturity in the digital and sustainable transformation including an example from one automotive company.

4 The Automotive Manufacturing Industry as a Twin Transformer

In order to meet the sustainability targets, like the Paris Agreement, the automotive industry is in a pivotal situation in which they must renovate itself to become more sustainable, achieve the decarbonization targets and fulfill SDGs [Ce23; LKB22; Pi21]. One of the tools to achieve the sustainability demands and remain competitive, as well as the other manufacturing sectors [Ce23], is through the use of digital technologies [ABA25; LRV21].

The opportunity for the automotive industry to become twin transformers has already called the attention of researchers, investigating industrial policy [TG23] and economic concepts [Ca24]. The latter was conducted as a case study, stressing that empirical evidence is necessary to validate the effectiveness of a twin transformation in the automotive sector. In order to continue with the research of twin transformation in the automotive sector, it is necessary to divide it between the product and the manufacturing (production processes), based on an increased product complexity and segmentate value chains, to increase the performance of a potential twin transformation [Ca24]. The manufacturing industry within the automotive sector represents a more attractive field of study since this one represents the basis of its business model [MF10]. Therefore, we propose to focus on the automotive manufacturing industry at first glance.

Aiming to answer our initial research question that intends to bring light to the twin transformation maturity of the automotive manufacturer industry, we focus our research efforts on German automotive manufacturer that employs hundreds of thousands of people with a significant global market share. The first author currently has a working relationship with the referred manufacturer, which could facilitate significant data access and analysis.

The following subsections outline our selected methodology and a preliminary problem analysis which illustrates our initial research steps and paves the road for our problem statement.

4.1 Methodology

Our research is steered by the recently developed eDSR approach outlined by [TWV24] which dissects design science research (DSR) projects in smaller self-contained echelons to facilitate its development, validation and communication in an independent form. Furthermore, it stresses an iterative development and demands rigorous evaluation for the creation of IS artifacts. The eDSR paradigm partitions the DSR in five echelons including (1) problem analysis, (2) objectives and requirements definition, (3) design and development, (4) demonstration and (5) evaluation [TWV24]. (1) Problem analysis is represented

by the problem statement [MH19] and validates the state of the art of the problem as well as its solvability [TWV24]. (2) Objectives and requirements definition validates them regarding applicability, coherence and completeness and is illustrated by the design objectives and requirements [Ma19; TWV24]. (3) Design and development aims to create a projectable design for functionality and its architecture evaluated through its applicability, consistency and elegance [TWV24]. (4) Demonstration highlights the instance of the artifact and should be assessed through a proof of concept [Nu15] via its ease of use, robustness and efficacy [TWV24]. (5) Evaluation is the final echelon and provides a contextualized artifact that fulfills the required consistency and completeness of the design [TWV24]. The validation techniques for each echelon will be selected at a later stage of our research taking into consideration the suggestions made by [TWV24]. At this stage, we select the validation techniques for the first echelon, problem analysis, described in the next section.

Additionally, we chose the qualitative method of case study, based on its popularity in the information systems research to study contemporary phenomenon in its real-world environment, in particular where the line between the phenomenon and its environment is ambiguous as it is the case of our research area [Re21].

4.2 Problem analysis (1st Echelon) – Preliminary problem statement

Starting with the first echelon, (1) problem analysis, we present an initial assessment of public and scientific information provided by our selected case study. As one of the biggest representatives of this industry, together with its European counterparts from Italy and France [Ca24], it aims to become greener in each of its manufacturing facilities developing ambitious strategies focused on the measurement of the absolute environmental impact of its manufacturing system [Ge23] and the strategy to achieve zero impact factories [Ge24]. Digitalization is part of the aspects of the “zero impact factory” strategy and it is described as follow: “Digitalization processes enable the transparent and optimized use of resources in factories through environmental data management from the machinery at the shopfloor level to the overall factory level.” [Ge24]. The content of digitalization within the strategy aligns with the ST-related capability for the operations dimension and therefore we can preliminarily conclude that our case organization fulfills at least the (1) maturity stage from the TTCMM [Br24] and potentially be considered as an ST expert.

At this stage, we would like to highlight that the previous conceptual analysis aims to represent preliminary results for the problem analysis since there is not information available about how in detail its strategy is implemented and how the digital and sustainability factors internally interact. We propose the consideration of the maturity models within the DesOps framework for manufacturing systems in the operations dimension of the TTCMM which simultaneously will contribute to the further corroboration and enhancement of the twin transformation. To support our proposal, we will continue our eDSR journey this year with the help of qualitative methods, initiating semi-structured interviews with sustainability and digitalization experts within our case study based on the literature

from the selected methodology [TWV24]. Additionally, we plan to collect additional data during the proposed case study aiming to assess factors such as strategy effectiveness and process dimensions between digitalization and sustainability. Our results will provide privileged knowledge about how the digitalization and sustainability efforts are orchestrated within their strategy and will guide our forthcoming echelons [Re21; TWV24].

The presented example aims to illustrate the need for further empirical research in the automotive manufacturer industry, emphasizing to explain how (adaptations of structures, processes, and culture) large organizations embark on their twin transformation journey, rather than exclusively describing what of their transformation endeavor. Multiple large automotive organizations have their own sustainability strategies [LKB22] and experience their own digital transformation [Ce23; LRV21], therefore they could also be investigated on within their ongoing twin transformation. As a result of its economic impact, digital-sustainable development and regionality, the study of the automotive manufacturing industry could provide valuable insights aiming to answer the current challenges the twin transformation presents.

5 Conclusion and future research

We investigated the potential maturity of the automotive manufacturing industry to become a twin transformer based on the presented eDSR and methodologies. Further, we plan to approach the in-depth investigation related to an empirical case study of the maturity stemming from automotive manufacturing systems by the hand of the problem statement in the first place.

Within our planned research plans, we strive to enhance the empirical research of the twin transformations, specifically focusing on the automotive manufacturing sector that later could be extended to broader industry levels and organizations (e.g., manufacturers and suppliers). Further, our research-in-progress hopes to benefit researchers and practitioners with practical and theoretical insights from this endeavor.

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